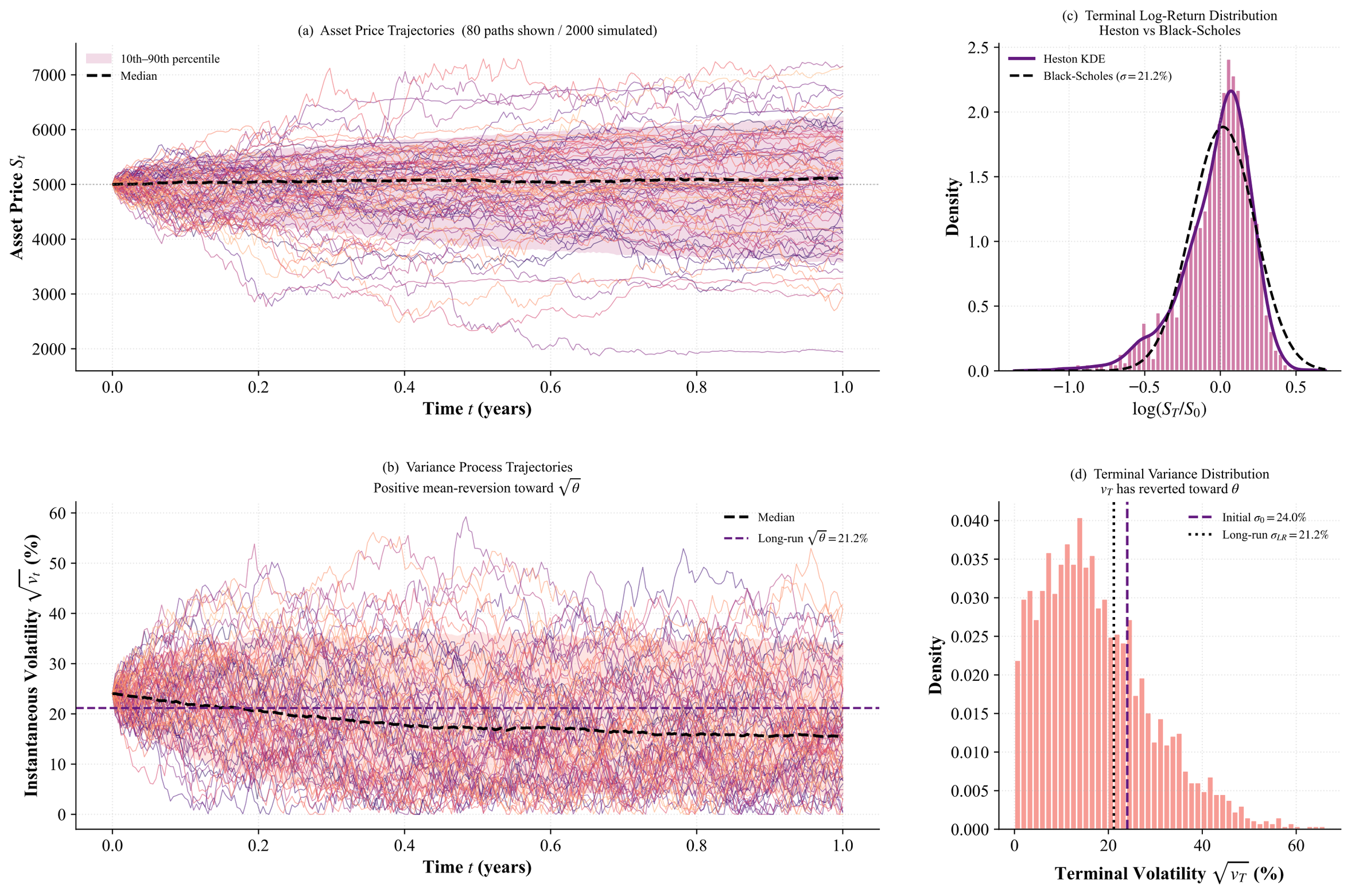
(a) Asset Price Trajectories (80 paths shown / 2000 simulated)
10th–90th percentile
Median
Asset Price S_t
Time t (years)
(b) Variance Process Trajectories
Positive mean-reversion toward $\sqrt{\theta}$
Median
Long-run $\sqrt{\theta} = 21.2\%$
Instantaneous Volatility $\sqrt{v_t}$ (%)
Time t (years)
(c) Terminal Log-Return Distribution
Heston vs Black-Scholes
Heston KDE
Black-Scholes ($\sigma = 21.2\%$)
Density
$\log(S_T/S_0)$
(d) Terminal Variance Distribution
v_T has reverted toward θ
Initial $\sigma_0 = 24.0\%$
Long-run $\sigma_{LR} = 21.2\%$
Density
Terminal Volatility $\sqrt{v_T}$ (%)